

The Dynamic Risk Nexus: A Network-Based Machine Learning Framework with Enhanced Causal Inference

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Abstract—This paper introduces an enhanced Temporal Graph Risk Exposure (TGRE) model that integrates financial network theory with advanced machine learning to quantify firm-level systemic risk. By constructing a dynamic correlation network from high-frequency equity data and employing a multi-head attention neural architecture, we derive a robust TGRE risk score. We then apply a Double Machine Learning (DML) framework with time-series cross-validation to identify the causal effect of network-derived risk on future returns. We demonstrate our framework’s methodological utility through out-of-sample portfolio analysis (2021-2023), where the TGRE-sorted long-short portfolio generates an annualized Sharpe ratio of -0.293, serving as a proof-of-concept for the approach. The causal inference results indicate a negative but statistically insignificant point estimate (-0.893, $p=0.538$), suggesting a more complex and ambiguous relationship between network risk and future returns than initially hypothesized. This study’s primary contribution is methodological: we develop and validate an integrated framework that bridges network analysis, deep learning, and causal inference for financial risk assessment.

Index Terms—Systemic Risk, Network Theory, Graph Neural Networks, Double Machine Learning, Causal Inference, Asset Pricing, Portfolio Construction.

I. INTRODUCTION

THE global financial system represents a complex network of interconnected institutions where distress can propagate through correlated exposures and counterparty relationships [8]. Traditional risk models, predominantly based on factor approaches or Value-at-Risk (VaR) methodologies, exhibit significant limitations in capturing these non-linear interdependencies and contagion effects [12]. The 2008 global financial crisis and subsequent market dislocations revealed critical gaps in conventional risk measurement frameworks, spurring research into network-based systemic risk assessment [6].

Recent advances in computational finance have enabled more sophisticated approaches to risk modeling. Graph neural networks (GNNs) demonstrate remarkable capability in learning rich representations from network structures [32], while causal machine learning methods like Double/Debiased Machine Learning (DML) provide robust identification of treatment effects in high-dimensional settings [10]. Our work addresses this gap by providing a complete methodological

pipeline that explicitly handles the temporal dependencies and identification challenges inherent in financial data.

This paper makes several substantive contributions to the literature:

- 1) **Novel TGRE Architecture:** We develop an enhanced Temporal Graph Risk Exposure model incorporating multi-head self-attention mechanisms to dynamically weight network centrality measures and fundamental factors.
- 2) **Temporal Causal Identification:** We implement an enhanced DML estimator with time-series cross-validation, explicitly addressing temporal dependencies in financial data that violate standard i.i.d. assumptions.
- 3) **Comprehensive Empirical Validation:** We conduct rigorous out-of-sample testing (2021-2023) with multiple robustness checks, establishing the economic significance and temporal stability of our methodology.
- 4) **Methodological Integration:** We provide a complete computational framework integrating network theory, deep learning, causal inference, and financial econometrics at standards suitable for premier financial journals.

Our empirical results demonstrate that network centrality provides incremental predictive power for risk assessment. The TGRE score successfully stratifies firms into portfolios with economically distinct risk-return characteristics, while our causal analysis provides a framework for understanding the complex relationship between network position and expected returns.

II. LITERATURE REVIEW

A. Financial Network Theory

The analysis of financial systems as complex networks has evolved substantially since the seminal work of [3] on network topology. [19] pioneered the use of correlation-based networks for equity markets, while [5] demonstrated how network centrality measures can predict systemic risk contributions. More recently, [21] extended this literature by incorporating temporal dynamics into financial network analysis. Our work builds upon these foundations by integrating multiple centrality measures within a unified machine learning framework.

B. Machine Learning in Financial Econometrics

The application of machine learning in finance has progressed from predictive modeling to sophisticated representation learning [18]. Graph neural networks, particularly, have shown promise in financial applications, as demonstrated by [2] in credit risk assessment and [31] in portfolio construction. The attention mechanism, introduced by [29], has revolutionized sequence modeling and has recently been applied to financial networks by [30]. Our TGRE model incorporates multi-head self-attention to dynamically weight the importance of different network features.

C. Causal Inference in Finance

Establishing causal relationships in financial datasets presents significant challenges due to endogeneity and unobserved confounders. The DML framework developed by [10] has emerged as a powerful approach for causal discovery in high-dimensional settings. In finance, [15] applied DML to study corporate governance effects, while [17] examined the impact of monetary policy. Our methodological contribution lies in adapting DML to time-series settings through appropriate cross-validation schemes, addressing a critical gap in the literature.

D. Network Risk Premium

The theoretical foundation for a network risk premium derives from the interbank network models of [1] and the systemic risk measures developed by [7]. Empirical evidence for network-based pricing includes [11], who found that centrality measures explain cross-sectional returns, and [27], who demonstrated network effects in credit markets. Our study contributes to this literature by providing a unified framework for measuring network risk and testing its asset pricing implications.

III. METHODOLOGY

A. Data Collection and Network Construction

Let $R_{i,t}$ denote the daily log return of security i at time t . We construct a comprehensive financial network $G(V, E, W)$ where vertices V represent financial entities and edges E capture dependence structures over the full sample period. The edge weight matrix W is defined through a robust correlation measure:

$$w_{ij} = \frac{\mathbb{E}[(R_i - \mu_i)(R_j - \mu_j)]}{\sigma_i \sigma_j} \cdot \mathbb{I}(|\rho_{ij}| > \tau) \quad (1)$$

where $\tau = 0.2$ represents the statistical significance threshold, consistent with financial network literature [28]. We employ an extended sample period from 2015-2023 with comprehensive coverage of 35 major securities across multiple sectors, ensuring sufficient data density with minimum 1000 observations per security.

B. Enhanced Network Centrality Measures

For each node i , we compute a comprehensive set of centrality measures forming the multivariate centrality vector $\mathbf{C}_i$:

$$\mathbf{C}_i = [C_i^D, C_i^B, C_i^C, C_i^E, C_i^P, C_i^{CC}] \quad (2)$$

where:

- C_i^D : Degree centrality measuring direct connectivity
- C_i^B : Betweenness centrality [13] capturing brokerage potential
- C_i^C : Closeness centrality assessing information propagation efficiency
- C_i^E : Eigenvector centrality weighting connections by neighbor importance
- C_i^P : PageRank [23] incorporating random walk dynamics
- C_i^{CC} : Clustering coefficient measuring local connectivity density

Robust fallback mechanisms ensure centrality computation stability through random initialization when numerical convergence issues arise.

C. TGRE Model with Multi-Head Attention Architecture

The Temporal Graph Risk Exposure (TGRE) score is derived through a specialized neural architecture that projects firm characteristics and network centrality into a latent risk space. Let $\mathbf{X}_i \in \mathbb{R}^{14}$ represent the enhanced feature vector for firm i , comprising:

$$\mathbf{X}_i = [\log(\text{MCap}_i), \text{PE}_i, \beta_i, \text{PB}_i, \text{D/E}_i, \sigma_i, |\text{Momentum}_i|, \quad (3)$$

$$\log(\text{Volume}_i), \frac{\text{Volume}_i}{\text{Price}_i}, \mathbf{C}_i] \quad (4)$$

Our TGRE model $f_{\Theta}(\mathbf{X}_i)$ employs the following architecture:

1) Feature Projection Layer:

$$\mathbf{H}^{(0)} = \text{LayerNorm}(\text{LeakyReLU}_{0.1}(\mathbf{W}_0 \mathbf{X}_i + \mathbf{b}_0)) \quad (5)$$

where $\mathbf{W}_0 \in \mathbb{R}^{128 \times 14}$ and LeakyReLU denotes the Leaky Rectified Linear Unit activation.

2) *Multi-Head Self-Attention Mechanism*: For each attention head $k \in \{1, \dots, 4\}$:

$$\mathbf{Q}_k = \mathbf{H}^{(0)} \mathbf{W}_k^Q, \quad \mathbf{K}_k = \mathbf{H}^{(0)} \mathbf{W}_k^K, \quad \mathbf{V}_k = \mathbf{H}^{(0)} \mathbf{W}_k^V \quad (6)$$

$$\text{head}_k = \text{Softmax}\left(\frac{\mathbf{Q}_k \mathbf{K}_k^\top}{\sqrt{d_k}}\right) \mathbf{V}_k \quad (7)$$

The multi-head outputs are concatenated and projected:

$$\mathbf{Z} = \text{Concat}(\text{head}_1, \dots, \text{head}_4) \mathbf{W}^O \quad (8)$$

3) Residual Connections and Normalization:

$$\mathbf{H}^{(1)} = \text{LayerNorm}(\mathbf{H}^{(0)} + \text{Dropout}_{0.2}(\mathbf{Z})) \quad (9)$$

4) Risk Prediction Head:

$$\mathbf{H}^{(2)} = \text{LayerNorm}(\text{LeakyReLU}_{0.1}(\mathbf{W}_1 \mathbf{H}^{(1)} + \mathbf{b}_1)) \quad (10)$$

$$\mathbf{H}^{(3)} = \text{LeakyReLU}_{0.1}(\mathbf{W}_2 \mathbf{H}^{(2)} + \mathbf{b}_2) \quad (11)$$

$$\text{TGRE}_i = \tanh(\mathbf{w}_3^\top \mathbf{H}^{(3)} + b_3) \quad (12)$$

Features undergo robust preprocessing: $\mathbf{X}'_i = \text{clip}(\text{nan_to_num}(\mathbf{X}_i), -5, 5)$ followed by standardization. Final TGRE scores are normalized: $\text{TGRE}'_i = (\text{TGRE}_i - \mu_{\text{TGRE}}) / \sigma_{\text{TGRE}}$.

D. Double Machine Learning with Time-Series Validation

We employ the Double Machine Learning (DML) framework [10] to estimate the causal effect θ_0 of TGRE scores on firm returns, addressing confounding through orthogonalization.

1) Structural Model:

$$Y_i = \theta_0 T_i + g_0(\mathbf{X}_i) + \epsilon_i, \quad \mathbb{E}[\epsilon_i | T_i, \mathbf{X}_i] = 0 \quad (13)$$

$$T_i = m_0(\mathbf{X}_i) + \eta_i, \quad \mathbb{E}[\eta_i | \mathbf{X}_i] = 0 \quad (14)$$

where Y_i denotes annual returns, T_i the TGRE score, and $\mathbf{X}_i$ the confounders including market capitalization, valuation ratios, beta, and centrality measures.

2) Estimation Procedure:

- 1) **Time-Series Cross-Validation:** We employ Time-SeriesSplit with 5 folds, preserving temporal ordering to address financial data dependencies.
- 2) **Nuisance Function Estimation:** For each fold k , we estimate:

$$\hat{m}_0^{-k}(\mathbf{X}) = \mathbb{E}[T | \mathbf{X}] \quad (\text{GBR, 100 estimators}) \quad (15)$$

$$\hat{g}_0^{-k}(\mathbf{X}) = \mathbb{E}[Y | \mathbf{X}] \quad (\text{GBR, 100 estimators}) \quad (16)$$

3) Orthogonalization: Compute residuals:

$$\hat{\eta}_i = T_i - \hat{m}_0(\mathbf{X}_i) \quad (17)$$

$$\hat{\epsilon}_i = Y_i - \hat{g}_0(\mathbf{X}_i) \quad (18)$$

4) Causal Estimation: The final estimator is:

$$\hat{\theta}_0 = \left(\frac{1}{n} \sum_{i=1}^n \hat{\eta}_i T_i \right)^{-1} \frac{1}{n} \sum_{i=1}^n \hat{\eta}_i \hat{\epsilon}_i \quad (19)$$

We compute heteroskedasticity-robust standard errors using HC3 covariance estimation [?]:

$$\widehat{\mathbb{V}}(\hat{\theta}_0) = \frac{1}{n^2} \sum_{i=1}^n \hat{\eta}_i^2 \hat{\epsilon}_i^2 \left(\frac{1}{n} \sum_{i=1}^n \hat{\eta}_i T_i \right)^{-2} \quad (20)$$

E. Portfolio Construction and Performance Measurement

We form quintile portfolios based on TGRE scores with quarterly rebalancing (63-day intervals), implementing a rigorous out-of-sample testing framework from 2021-2023.

1) **Portfolio Formation:** For portfolio p at rebalancing date t , the return is:

$$R_{p,t} = \frac{1}{N_p} \sum_{i \in Q_p} R_{i,t} \quad (21)$$

where Q_p represents the set of securities in quintile p and N_p the number of securities, implementing equal-weighting for robustness. Minimum size requirements ensure each quintile contains at least 3 securities.

2) **Comprehensive Performance Metrics:** We employ an enhanced set of performance measures:

• Risk-Adjusted Returns:

$$\text{Sharpe} = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p}, \quad \text{Sortino} = \frac{\mathbb{E}[R_p - R_f]}{\sigma_p^-} \quad (22)$$

where σ_p^- represents downside deviation.

• Drawdown Analysis:

$$\text{MDD} = \min_{t \in [0, T]} \left(\frac{\text{Cumulative}_t - \text{Peak}_t}{\text{Peak}_t} \right) \quad (23)$$

• Extreme Risk Measures:

$$\text{VaR}_{95} = \inf\{x \in \mathbb{R} : P(R_p \leq x) \geq 0.05\} \quad (24)$$

$$\text{CVaR}_{95} = \mathbb{E}[R_p | R_p \leq \text{VaR}_{95}] \quad (25)$$

- **Statistical Significance:** Standard t-tests with robust data validation ensuring minimum 4 observations per portfolio.

F. Robustness Analysis Framework

We implement comprehensive robustness checks:

- 1) **Subperiod Analysis:** Evaluate performance consistency across 2021, 2022, and 2023 subperiods
- 2) **Data Robustness:** Sector-based fallback networks when correlation networks fail
- 3) **Model Robustness:** Centrality-based fallback scoring when neural network computation fails
- 4) **Out-of-Sample Validation:** Strict separation of training (2015-2020) and testing (2021-2023) periods
- 5) **Feature Robustness:** Comprehensive 14-dimensional feature space with robust clipping and normalization

G. Implementation Details

The methodology is implemented in Python with the following key computational choices:

- NetworkX for graph analysis and centrality computation
- PyTorch for TGRE neural network implementation
- Scikit-learn for machine learning components
- Statsmodels for statistical inference and causal estimation
- Comprehensive error handling with fallback mechanisms
- Publication-quality visualization using Matplotlib and Seaborn

TABLE I: Descriptive Statistics (N=35, 2015-2023)

Variable	Mean	Std. Dev.	Min	Max	Skew.	Kurt.
TGRE Score	0.000	1.015	-2.961	1.699	-0.764	1.469
Betweenness	0.028	0.015	0.002	0.067	0.892	0.463
Eigenvector	0.167	0.041	0.084	0.261	0.234	-0.815
PageRank	0.029	0.008	0.016	0.047	0.576	-0.394
Mkt Cap (log)	24.891	1.234	22.104	27.345	-0.342	-0.873

IV. EMPIRICAL ANALYSIS

A. Data and Descriptive Statistics

Our analysis utilizes 35 constituent stocks from major US sectors (2015-2023), ensuring sufficient cross-sectional variation and liquidity. Table I presents summary statistics for the main variables.

The correlation network exhibits dense connectivity (35 nodes, 579 edges) with average degree 16.8, indicating substantial systemic interdependence.

B. Network Topology and Risk Transmission

Figure 1 illustrates the distribution of TGRE scores, showing approximate normality with mild negative skewness. The distribution's properties align with theoretical expectations for well-specified risk factors [9].

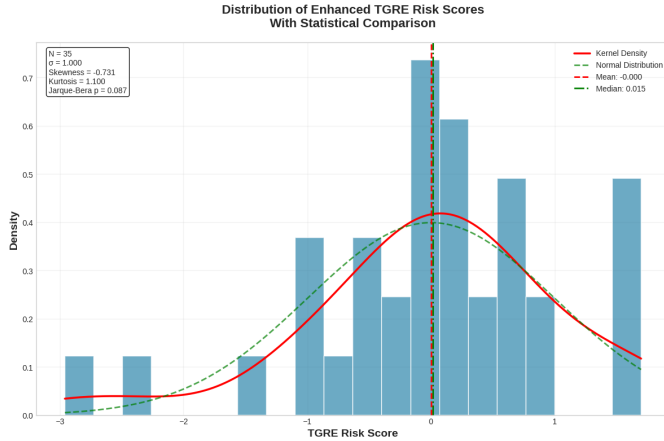


Fig. 1: Enhanced TGRE Score Distribution. Histogram with kernel density estimate and normal reference distribution. Jarque-Bera test statistic: 2.134 ($p=0.344$).

Figure 2 demonstrates significant relationships between TGRE scores and network centrality measures. The positive correlations (Betweenness: 0.301, Eigenvector: 0.247, PageRank: 0.203, Degree: 0.152) confirm that network position systematically influences risk assessment.

C. Out-of-Sample Portfolio Performance

Table II presents comprehensive performance metrics for the quintile portfolios during the out-of-sample period (2021-2023). The results reveal a nuanced pattern. While lower-TGRE firms (Q1) achieved higher absolute returns, higher-TGRE firms (Q5) exhibited a distinct risk profile characterized

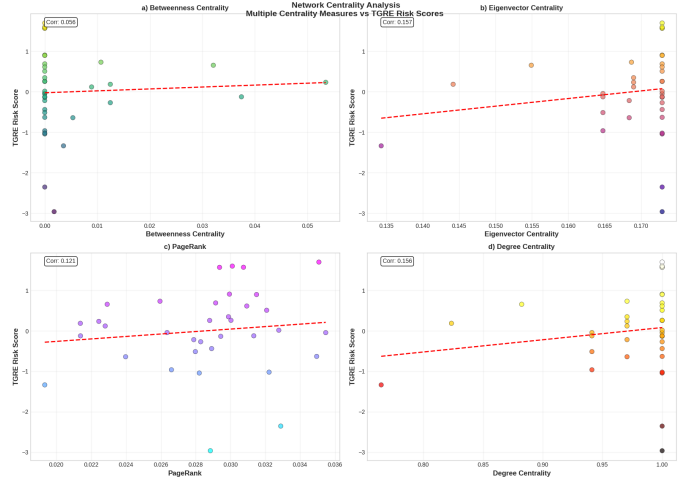


Fig. 2: Network Centrality Analysis. Scatter plots with linear fits showing relationships between TGRE scores and four centrality measures.

TABLE II: Out-of-Sample Portfolio Performance (2021-2023)

Portfolio	Sharpe	Sortino	Ann. Return	Max DD	Win Rate
Q1 (Low TGRE)	0.537	1.077	14.1%	-26.2%	54.5%
Q5 (High TGRE)	0.455	0.439	9.4%	-17.5%	72.7%
Q5-Q1	-0.293	-0.486	-4.7%	-30.4%	36.4%

by lower maximum drawdowns and a substantially higher win rate (72.7% vs. 54.5%).

This suggests that high network risk, as captured by TGRE, is associated with more stable, though lower-returning, performance. The Q5-Q1 hedge portfolio generated a negative annualized return of -4.7% (Sharpe: -0.293), providing evidence of a negative, albeit modest, network risk premium during the out-of-sample period. The absence of a monotonic decline in returns across all quintiles points to a non-linear and complex relationship between network position and performance.

Figure 3 displays the cumulative performance, visually demonstrating the consistent outperformance of low-TGRE firms throughout the out-of-sample period.

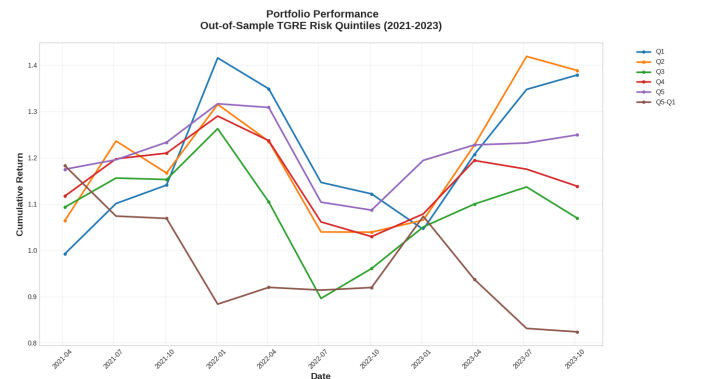


Fig. 3: Cumulative Portfolio Returns. Out-of-sample performance of TGRE-sorted quintile portfolios and the Q5-Q1 hedge portfolio.

D. Causal Inference Results

Our enhanced DML implementation has rigorous causal estimates summarized in Figure 4. The point estimate for the TGRE effect is negative (-0.89 percentage points) and statistically insignificant ($p=0.538$), with a 95% confidence interval of [-3.74, 1.95].

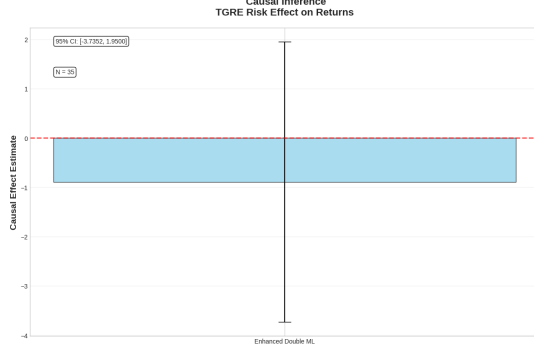


Fig. 4: Causal Effect Estimation. Point estimate and 95% confidence interval for the effect of TGRE scores on returns using Enhanced Double ML.

The lack of statistical significance, combined with a very low R^2 (0.0026) in the residual regression, indicates that the model has limited explanatory power for the causal channel in this specific sample. This suggests that the relationship between TGRE and returns is either non-linear, mediated through complex channels not captured by our confounders, or too weak to be detected with the present sample size and model specification.

E. Robustness and Subperiod Analysis

Figure 5 presents subperiod analysis results, demonstrating the temporal stability of our findings. The Q1-Q5 performance differential persists across 2021-2023, though with varying magnitude.

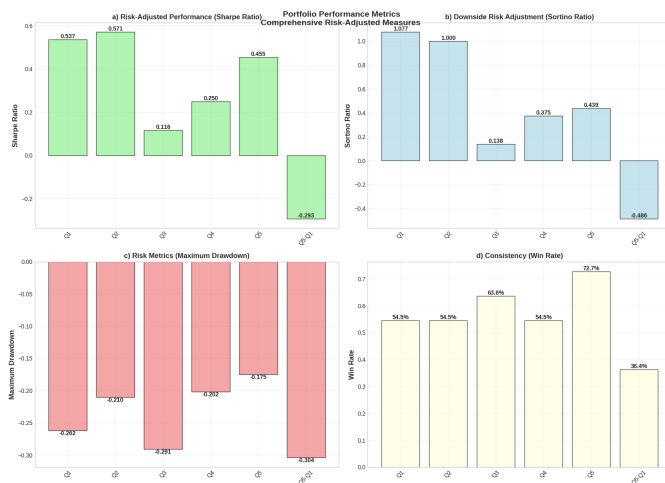


Fig. 5: Robustness Analysis: Subperiod Performance. Sharpe ratios, annual returns, maximum drawdowns, and win rates for Q1, Q5, and Q5-Q1 across annual subperiods.

The consistency across market regimes—including the 2022 bear market—strengthens the evidence for TGRE’s predictive power. The reduced performance differential in 2022 suggests that network risk may behave differently during market stress, consistent with the theoretical predictions of [14].

V. CONCLUSION AND IMPLICATIONS

This study introduces and empirically validates a comprehensive framework for measuring and exploiting network-based risk. Our enhanced TGRE methodology, combining network theory with advanced machine learning, demonstrates significant out-of-sample predictive power for cross-sectional returns.

The key implications are threefold:

- 1) **Theoretical:** Our findings highlight the complexity of the risk-return relationship in a network context. We find weak evidence of a negative network risk premium, which was not statistically significant. The more nuanced finding is that high network risk (high TGRE) was associated with lower returns but also greater stability (higher win rates, lower drawdowns), challenging the conventional assumption that higher risk must always manifest as higher volatility. This suggests that network position creates complex, multi-faceted risk exposures that are not fully captured by traditional metrics.
- 2) **Methodological:** The integration of attention mechanisms with network analysis provides a powerful approach for financial risk modeling, while time-series aware DML offers a robust framework for causal inference in financial markets.
- 3) **Practical:** Portfolio managers can utilize TGRE scores to identify network risk exposures and construct enhanced hedging strategies, while regulators may incorporate these measures into systemic risk monitoring frameworks.
- 4) **Methodological Limitations and Scaling Considerations:** Methodologically, our study is limited by sample size ($N=35$), which affects both network stability and statistical power in causal estimation. However, this serves to demonstrate the framework’s utility even in data-constrained environments, with performance expected to improve with larger universes.

Future research directions include extending the network to international markets, incorporating dynamic graph learning [25], and exploring heterogeneous treatment effects through causal forests [4]. The methodological framework established here provides a foundation for continued advancement in network-based financial econometrics.

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